

Project – Dealing with time series

Seunghyoung Ryu

세종대학교 AI로봇학과

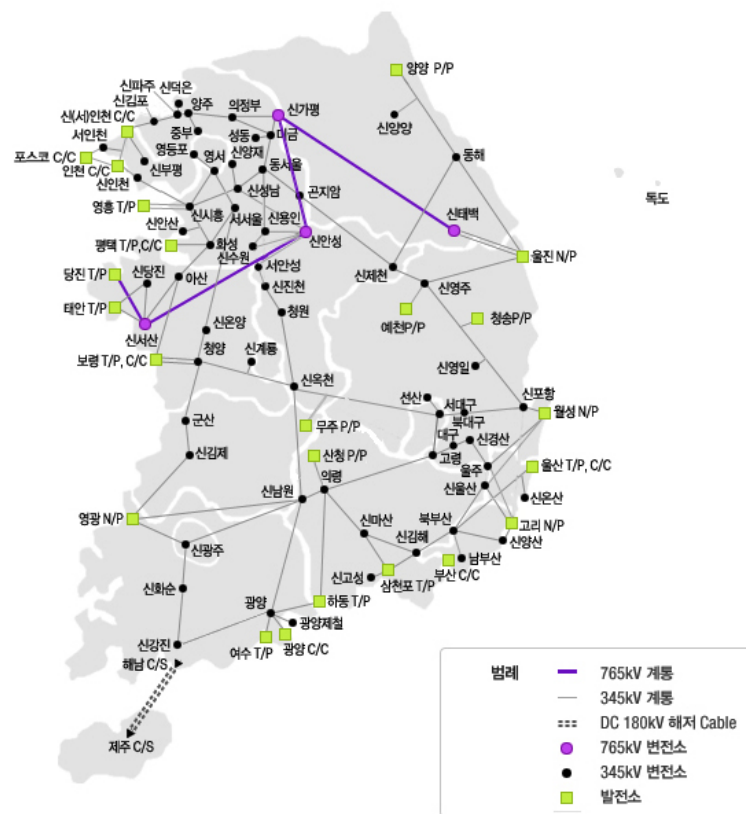
2026 – 1

Outline

- Data
- Target
- Process

Electricity Transformer

- Transformer
- Convert voltage



Electricity Transformer Dataset (ETDataset)

- ETT-small: The data of 2 Electricity Transformers at 2 stations, including load, oil temperature.
- (we will use one of them)
- Zhou, Haoyi, et al. "Informer: Beyond efficient transformer for long sequence time-series forecasting." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 35. No. 12. 2021.

Dataset Overview

We use the `.csv` file format to save the data, a demo of the ETT-small data is illustrated in Figure 3. The first line (8 columns) is the horizontal header and includes "date", "HUFL", "HULL", "MUFL", "MULL", "LUFL", "LULL" and "OT". The detailed meaning of each column name is shown in the Table 1.

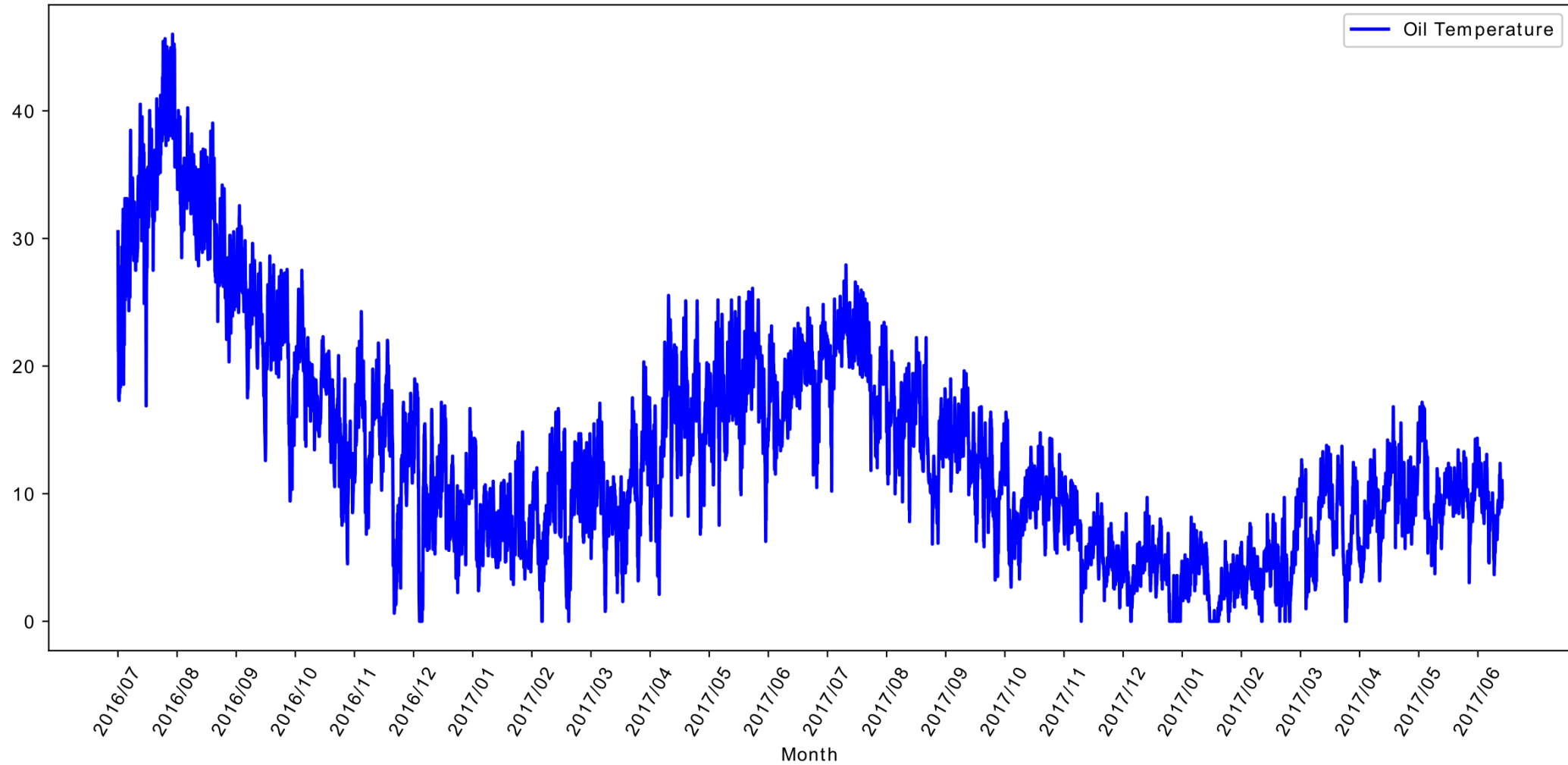
	date	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
0	2016-07-01 00:00:00	5.827	2.009	1.599	0.462	4.203	1.340	30.531000
1	2016-07-01 00:15:00	5.760	2.076	1.492	0.426	4.264	1.401	30.459999
2	2016-07-01 00:30:00	5.760	1.942	1.492	0.391	4.234	1.310	30.038000
3	2016-07-01 00:45:00	5.760	1.942	1.492	0.426	4.234	1.310	27.013000
4	2016-07-01 01:00:00	5.693	2.076	1.492	0.426	4.142	1.371	27.787001

Figure 3. A demo of the ETT data.

Field	date	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
Description	The recorded date	High UseFul Load	High UseLess Load	Middle UseFul Load	Middle UseLess Load	Low UseFul Load	Low UseLess Load	Oil Temperature (target)

Table 1. Description for each column.

Dataset Overview



Dataset overview

- Target: OT (Oil Temperature)
- 2016-07-01 00:00:00 ~ 2018-06-26 19:00:00

date	HUFL	HULL	MUFL	MULL	LUFL	LULL	OT
2016-07-01 00:00:00	5.827000141143800	2.009000062942510	1.5989999771118200	0.4620000123977660	4.203000068664550	1.3400000333786000	30.5310001373291
2016-07-01 01:00:00	5.692999839782720	2.075999975204470	1.4919999837875400	0.4259999990463260	4.142000198364260	1.371000051498410	27.78700065612790
2016-07-01 02:00:00	5.1570000648498500	1.741000056266790	1.2790000438690200	0.35499998927116400	3.776999950408940	1.218000054359440	27.78700065612790
2016-07-01 03:00:00	5.0900001525878900	1.9420000314712500	1.2790000438690200	0.3910000026226040	3.806999921798710	1.2790000438690200	25.04400062561040
2016-07-01 04:00:00	5.357999801635740	1.9420000314712500	1.4919999837875400	0.4620000123977660	3.868000030517580	1.2790000438690200	21.947999954223600
2016-07-01 05:00:00	5.625999927520750	2.1429998874664300	1.5279999971389800	0.5329999923706060	4.051000118255620	1.371000051498410	21.173999786377000
2016-07-01 06:00:00	7.166999816894530	2.9470000267028800	2.131999969482420	0.7820000052452090	5.026000022888180	1.8580000400543200	22.79199981689450
2016-07-01 07:00:00	7.434999942779540	3.282000064849850	2.309999942779540	1.031000018119810	5.086999893188480	2.2239999771118200	23.143999099731400

Project

- Train/Val/Test : 60:20:20 (Build your own training/validation set)
- 2016-07-01 00:00:00 ~ 2018-06-26 19:00:00
- Test : 2018. 02.01 ~ 2018.06.22
- Objective : Predict OT of next 4 days. (d = 96)

ID	T0	T1	T2	...	T95
2018.02.01	00:00+0h	00:00+1h	00:00+2h		00:00+95h
2018.02.01	00:00+0h				
...					
2018.06.22	00:00+0h				00:00+95h

Input example

- Output is 96 dimensional vector
- Input is $L \times (7+k)$ matrix
 - 5: HUFL, HULL, MUFL, MULL, LUFL, LULL, OT
 - k: create your own features.
 - L: select on your own.
- e.g., If $k=5$, $L=4$
 - target date : 2018.02.01
 - features from 2018.01.31 – [20:00, 21:00, 22:00, 23:00]

Outcome

- Do coding (data analysis, model development)
- Submit your prediction on Kaggle
- Project report: 5~10 page (no predefined format)
 - You could use Kaggle notebook.
- Submit your report on online campus website.
- Public score and private score will be given.

Important dates

- (Kaggle submission) 6/5, 12pm
- (Report submission) 6/9, 12pm
- (Presentation) 6/10, top-5 & volunteers (3 students)

Kaggle Link

- <https://www.kaggle.com/t/8cb454699ac145a6aa148f8c4687d923>
- Note: Maximum daily submission limit is 5.